

TOM GUO

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EDUCATION

- Expected 12/27 **NEW YORK UNIVERSITY** New York, NY
The Courant Institute School of Mathematics, Computing, and Data Science
M.S. in Mathematics in Finance
- **Expected Coursework:** stochastic calculus, dynamic asset pricing, derivatives modeling, statistical machine learning, computational finance, algorithmic trading systems
- 09/21 - 06/26 **UNIVERSITY OF WATERLOO** Waterloo, ON
Bachelors in Mathematics
Double Major in Mathematical Finance and Statistics, Minor in Computing
- **Coursework:** measure-theoretic probability, stochastic processes, quantitative risk management, time series forecasting, numerical PDEs, Monte Carlo simulation, optimization in finance, derivative pricing, data structure and algorithms, object-oriented programming
 - **Honors/Awards:** NSERC undergraduate research award

EXPERIENCE

- 05/25 - 08/25 **ONEX CORPORATION** (\$56B AUM, TSX: ONEX) Toronto, ON
Quantitative Risk Management Intern (Python, SQL)
- Monitored fund and portfolio-level exposures using Bloomberg, analyzing VaR, beta, and sector concentration to guide investment decisions and strengthen portfolio risk management strategies
 - Conducted equity research using Bloomberg, building automated screening tools to identify high-performing metrics across global equities and support systematic idea generation
 - Designed Python-based quantitative models to analyze relative value, historical trends, and cross-sectional market data, translating data-driven insights into actionable investment strategies
 - Implemented scenario analysis and stress-testing frameworks to quantify potential performance and assess portfolio resilience under macroeconomic shocks
- 01/25 - 05/25 **ERGO NEXT INSURANCE** (Acquired by Munich Re) Palo Alto, CA
Reserving Actuary (Python, SQL)
- Improved efficiency and accuracy in loss ratio calculations and reserve forecasts by automating reserving workflows with SQL, Python, and Excel macros, streamlining data processing
 - Developed Python/SQL tools for claim pulls and development factor interpolation, enhancing efficiency of pricing and financial forecasting initiatives across actuarial and data science teams
- 05/23 - 08/24 **MCMASTER UNIVERSITY** (Top Health Science Research in Canada) Hamilton, ON
Machine Learning Research Intern (Python, PyTorch)
- Researched and implemented advanced machine learning methods to analyze hundreds of retinal images, emphasizing statistical pattern recognition and predictive modeling for disease diagnosis
 - Designed and deployed a Python/PyTorch pipeline that standardized image quality, increasing dataset reliability and improving diagnostic model accuracy by 30%

PROJECT

- 05/25 - 12/25 **UNIVERSITY OF WATERLOO** Waterloo, ON
CNN Crypto Imaging Research (Python, ML)
- Forecasted cryptocurrency price trends with convolutional neural network models, drawing on methods from “(Re-)Imag(in)ing Price Trends,” applying research to practical market situations
 - Classified directional price movements on rolling quarterly basis using historical OHLC return data, extracting predictive signals, and evaluating their statistical validity
 - Developed and backtested systematic trading strategies informed by model outputs, assessing profitability, risk-adjusted returns, and practical applicability in real-market context

COMPUTATIONAL SKILLS / OTHER

Programming Languages: Python, C, SQL, R, Typescript

Languages: English (native), Mandarin (fluent)

Certifications: Bloomberg Market Concepts, Bloomberg Finance Fundamentals